```
  ___  ____  ____  ____  ____(R)
 /__    /   ____/   /   ____/
___/   /   /___/   /   /___/
  Statistics/Data analysis
```

1 . do "C:\Users\fahim\AppData\Local\Temp\STDaf88_000000.tmp"

2 .

3 . anova StrokeVolume i.TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)

Number of obs =	**52**	R-squared =	**0.7788**
Root MSE =	**10.3143**	Adj R-squared =	**0.6239**

Source	Partial SS	df	MS	F	Prob>F
Model	**11234.69**	**21**	**534.98526**	**5.03**	**0.0000**
TimeNumer~l	**922.07546**	**2**	**461.03773**	**4.33**	**0.0222**
GroupNume~l	**506.24084**	**2**	**253.12042**	**2.38**	**0.1099**
TimeNumer~l#GroupNume~l	**766.10565**	**4**	**191.52641**	**1.80**	**0.1549**
Sex	**136.4315**	**1**	**136.4315**	**1.28**	**0.2664**
weight	**7825.882**	**12**	**652.15684**	**6.13**	**0.0000**
Residual	**3191.5405**	**30**	**106.38468**		
Total	**14426.231**	**51**	**282.86727**		

4 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	**2**	**4.33**	**0.0222**
1	**2**	**3.55**	**0.0414**
2	**2**	**0.47**	**0.6291**
Joint	**6**	**2.78**	**0.0284**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-5.989356**	**5.954961**	**-1.01**	**0.323**	**-18.15101**	**6.172297**
(30 vs base) 1	**-7.774981**	**6.385984**	**-1.22**	**0.233**	**-20.8169**	**5.266938**
(30 vs base) 2	**-3.710964**	**5.954961**	**-0.62**	**0.538**	**-15.87262**	**8.450689**
(74 vs base) 0	**11.27486**	**5.954961**	**1.89**	**0.068**	**-.8867971**	**23.43651**
(74 vs base) 1	**9.098544**	**5.954961**	**1.53**	**0.137**	**-3.063109**	**21.2602**
(74 vs base) 2	**-6.056512**	**6.385984**	**-0.95**	**0.351**	**-19.09843**	**6.985408**

```
5 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	**2**	**2.59**	**0.0913**
30	**2**	**2.87**	**0.0723**
74	**2**	**2.73**	**0.0817**
Joint	**6**	**2.21**	**0.0695**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-27.3988**	**14.19647**	**-1.93**	**0.063**	**-56.39187**	**1.594265**
(1 vs base) 30	**-29.18443**	**14.98632**	**-1.95**	**0.061**	**-59.79058**	**1.421727**
(1 vs base) 74	**-29.57511**	**14.19647**	**-2.08**	**0.046**	**-58.56818**	**-.5820466**
(2 vs base) 0	**-2.497274**	**9.754752**	**-0.26**	**0.800**	**-22.41914**	**17.42459**
(2 vs base) 30	**-.2188824**	**9.754752**	**-0.02**	**0.982**	**-20.14074**	**19.70298**
(2 vs base) 74	**-19.82864**	**9.845215**	**-2.01**	**0.053**	**-39.93525**	**.2779706**

```
6 .
7 .
8 . anova StrokeWork TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)
```

Number of obs =	**52**	R-squared =	**0.9014**
Root MSE =	**1059.34**	Adj R-squared =	**0.8324**

Source	Partial SS	df	MS	F	Prob>F
Model	**3.078e+08**	**21**	**14658406**	**13.06**	**0.0000**
TimeNumer~l	**1.676e+08**	**2**	**83808542**	**74.68**	**0.0000**
GroupNume~l	**110635.26**	**2**	**55317.629**	**0.05**	**0.9520**
TimeNumer~l#GroupNume~l	**54580435**	**4**	**13645109**	**12.16**	**0.0000**
Sex	**73958.627**	**1**	**73958.627**	**0.07**	**0.7991**
weight	**37548629**	**12**	**3129052.4**	**2.79**	**0.0112**
Residual	**33665906**	**30**	**1122196.9**		
Total	**3.415e+08**	**51**	**6695930**		

```
9 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	**2**	**10.67**	**0.0003**
1	**2**	**78.21**	**0.0000**
2	**2**	**10.37**	**0.0004**
Joint	**6**	**33.08**	**0.0000**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-1849.289**	**611.609**	**-3.02**	**0.005**	**-3098.362**	**-600.2171**
(30 vs base) 1	**-2020.25**	**655.8776**	**-3.08**	**0.004**	**-3359.731**	**-680.7696**
(30 vs base) 2	**-1360.784**	**611.609**	**-2.22**	**0.034**	**-2609.856**	**-111.7118**
(74 vs base) 0	**925.639**	**611.609**	**1.51**	**0.141**	**-323.4333**	**2174.711**
(74 vs base) 1	**5679.985**	**611.609**	**9.29**	**0.000**	**4430.913**	**6929.057**
(74 vs base) 2	**1624.569**	**655.8776**	**2.48**	**0.019**	**285.0885**	**2964.05**

```
10 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	**2**	**0.32**	**0.7316**
30	**2**	**0.70**	**0.5063**
74	**2**	**3.77**	**0.0345**
Joint	**6**	**8.14**	**0.0000**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-1115.531**	**1458.06**	**-0.77**	**0.450**	**-4093.288**	**1862.225**
(1 vs base) 30	**-1286.492**	**1539.182**	**-0.84**	**0.410**	**-4429.922**	**1856.937**
(1 vs base) 74	**3638.814**	**1458.06**	**2.50**	**0.018**	**661.0582**	**6616.571**
(2 vs base) 0	**-309.0912**	**1001.87**	**-0.31**	**0.760**	**-2355.182**	**1737**
(2 vs base) 30	**179.4141**	**1001.87**	**0.18**	**0.859**	**-1866.677**	**2225.505**
(2 vs base) 74	**389.8391**	**1011.161**	**0.39**	**0.703**	**-1675.227**	**2454.905**

```
11 .
12 .
13 . anova EjectionFraction TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)

                         Number of obs =         52    R-squared     =  0.7856
                         Root MSE      =    7.53489    Adj R-squared =  0.6356

                  Source | Partial SS         df         MS        F    Prob>F
    ---------------------+----------------------------------------------------
                   Model |  6242.5873         21   297.26606      5.24  0.0000
                         |
            TimeNumer~l  |  2744.5011          2   1372.2505     24.17  0.0000
            GroupNume~l  |  84.586524          2   42.293262      0.74  0.4833
 TimeNumer~l#GroupNume~l |  567.45518          4    141.8638      2.50  0.0636
                     Sex |  2.8040872          1   2.8040872      0.05  0.8256
                  weight |    2476.72         12   206.39333      3.64  0.0020
                         |
                Residual |   1703.235         30   56.774501
    ---------------------+----------------------------------------------------
                   Total |  7945.8223         51   155.80044

14 . contrast TimeNumerical@GroupNumerical , effect

   Contrasts of marginal linear predictions

   Margins: asbalanced

   ------------------------------------------------
                               |         df          F        P>F
   ----------------------------+-------------------------------------
   TimeNumerical@GroupNumerical |
                             0 |          2       7.82     0.0018
                             1 |          2       6.68     0.0040
                             2 |          2      14.41     0.0000
                         Joint |          6       9.64     0.0000
                               |
                   Denominator |         30
   ------------------------------------------------

   ---------------------------------------------------------------------------------------
                               |   Contrast   Std. err.      t    P>|t|     [95% conf. interval]
   ----------------------------+----------------------------------------------------------
   TimeNumerical@GroupNumerical |
                (30 vs base) 0 |   16.81036   4.350268     3.86   0.001     7.925929    25.69479
                (30 vs base) 1 |   16.41327   4.665143     3.52   0.001     6.885778    25.94076
                (30 vs base) 2 |   17.68911   4.350268     4.07   0.000     8.804675    26.57354
                (74 vs base) 0 |   11.57262   4.350268     2.66   0.012     2.688187    20.45705
                (74 vs base) 1 |   3.322994   4.350268     0.76   0.451    -5.561439    12.20743
                (74 vs base) 2 |  -5.616267   4.665143    -1.20   0.238    -15.14376    3.911226
   ---------------------------------------------------------------------------------------
```

15 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	**2**	**1.39**	**0.2647**
30	**2**	**1.54**	**0.2314**
74	**2**	**1.31**	**0.2851**
Joint	**6**	**2.11**	**0.0811**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-5.533987**	**10.37093**	**-0.53**	**0.598**	**-26.71425**	**15.64627**
(1 vs base) 30	**-5.931078**	**10.94794**	**-0.54**	**0.592**	**-28.28974**	**16.42759**
(1 vs base) 74	**-13.78361**	**10.37093**	**-1.33**	**0.194**	**-34.96387**	**7.396647**
(2 vs base) 0	**6.434912**	**7.126124**	**0.90**	**0.374**	**-8.118574**	**20.9884**
(2 vs base) 30	**7.313658**	**7.126124**	**1.03**	**0.313**	**-7.239827**	**21.86714**
(2 vs base) 74	**-10.75398**	**7.19221**	**-1.50**	**0.145**	**-25.44243**	**3.934476**

16 .

17 . anova HeartRate TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)

Number of obs = **52**   R-squared = **0.7805**
Root MSE = **20.3148**   Adj R-squared = **0.6269**

Source	Partial SS	df	MS	F	Prob>F
Model	**44034.457**	**21**	**2096.8789**	**5.08**	**0.0000**
TimeNumer~l	**28143.113**	**2**	**14071.556**	**34.10**	**0.0000**
GroupNume~l	**683.34787**	**2**	**341.67393**	**0.83**	**0.4467**
TimeNumer~l#GroupNume~l	**3432.412**	**4**	**858.103**	**2.08**	**0.1084**
Sex	**1420.385**	**1**	**1420.385**	**3.44**	**0.0734**
weight	**10634.581**	**12**	**886.21507**	**2.15**	**0.0443**
Residual	**12380.7**	**30**	**412.69**		
Total	**56415.157**	**51**	**1106.1795**		

18 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	**2**	**8.36**	**0.0013**
1	**2**	**20.22**	**0.0000**
2	**2**	**10.84**	**0.0003**
Joint	**6**	**13.14**	**0.0000**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	**36.41454**	**11.72874**	**3.10**	**0.004**	**12.46126**	**60.36782**
(30 vs base) 1	**25.57948**	**12.57767**	**2.03**	**0.051**	**-.1075496**	**51.26652**
(30 vs base) 2	**46.921**	**11.72874**	**4.00**	**0.000**	**22.96771**	**70.87428**
(74 vs base) 0	**45.22415**	**11.72874**	**3.86**	**0.001**	**21.27087**	**69.17743**
(74 vs base) 1	**73.64222**	**11.72874**	**6.28**	**0.000**	**49.68893**	**97.5955**
(74 vs base) 2	**49.94643**	**12.57767**	**3.97**	**0.000**	**24.2594**	**75.63346**

19 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	**2**	**0.72**	**0.4963**
30	**2**	**2.11**	**0.1387**
74	**2**	**0.06**	**0.9416**
Joint	**6**	**1.62**	**0.1768**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-24.32657**	**27.96101**	**-0.87**	**0.391**	**-81.43058**	**32.77744**
(1 vs base) 30	**-35.16163**	**29.51668**	**-1.19**	**0.243**	**-95.44273**	**25.11948**
(1 vs base) 74	**4.0915**	**27.96101**	**0.15**	**0.885**	**-53.01251**	**61.19551**
(2 vs base) 0	**1.95113**	**19.21271**	**0.10**	**0.920**	**-37.28646**	**41.18872**
(2 vs base) 30	**12.45758**	**19.21271**	**0.65**	**0.522**	**-26.78001**	**51.69518**
(2 vs base) 74	**6.673411**	**19.39089**	**0.34**	**0.733**	**-32.92806**	**46.27488**

```
20 .
21 .
22 . anova CardiacOutput TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)
```

Number of obs = 52 R-squared = 0.8157
Root MSE = 1715.14 Adj R-squared = 0.6867

Source	Partial SS	df	MS	F	Prob>F
Model	3.906e+08	21	18600226	6.32	0.0000
TimeNumer~l	95502961	2	47751481	16.23	0.0000
GroupNume~l	13161460	2	6580730.1	2.24	0.1243
TimeNumer~l#GroupNume~l	37068877	4	9267219.1	3.15	0.0282
Sex	8845115.2	1	8845115.2	3.01	0.0932
weight	2.178e+08	12	18148049	6.17	0.0000
Residual	88251646	30	2941721.5		
Total	4.789e+08	51	9389340.9		

```
23 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	2	7.04	0.0031
1	2	15.51	0.0000
2	2	1.35	0.2738
Joint	6	7.97	0.0000
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	1079.114	990.2393	1.09	0.284	-943.2246	3101.452
(30 vs base) 1	160.1118	1061.913	0.15	0.881	-2008.605	2328.828
(30 vs base) 2	1560.822	990.2393	1.58	0.125	-461.5163	3583.16
(74 vs base) 0	3618.327	990.2393	3.65	0.001	1595.989	5640.665
(74 vs base) 1	4946.694	990.2393	5.00	0.000	2924.356	6969.033
(74 vs base) 2	1222.735	1061.913	1.15	0.259	-945.9816	3391.451

```
24 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	2	2.83	0.0746
30	2	4.10	0.0267
74	2	1.43	0.2547
Joint	6	2.98	0.0210
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	-4559.934	2360.705	-1.93	0.063	-9381.137	261.2686
(1 vs base) 30	-5478.936	2492.048	-2.20	0.036	-10568.38	-389.4961
(1 vs base) 74	-3231.567	2360.705	-1.37	0.181	-8052.77	1589.636
(2 vs base) 0	-186.8545	1622.099	-0.12	0.909	-3499.623	3125.914
(2 vs base) 30	294.8537	1622.099	0.18	0.857	-3017.915	3607.623
(2 vs base) 74	-2582.447	1637.142	-1.58	0.125	-5925.938	761.044

```
25 .
26 . anova Elastance TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)
```

Number of obs =	52	R-squared =	0.8289
Root MSE =	.721371	Adj R-squared =	0.7091

Source	Partial SS	df	MS	F	Prob>F
Model	75.619653	21	3.6009359	6.92	0.0000
TimeNumer~l	30.906463	2	15.453232	29.70	0.0000
GroupNume~l	3.0575336	2	1.5287668	2.94	0.0684
TimeNumer~l#GroupNume~l	15.966197	4	3.9915492	7.67	0.0002
Sex	.31889229	1	.31889229	0.61	0.4399
weight	15.19028	12	1.2658567	2.43	0.0239
Residual	15.611277	30	.52037589		
Total	91.23093	51	1.7888418		

```
27 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	**2**	**1.60**	**0.2179**
1	**2**	**22.26**	**0.0000**
2	**2**	**19.41**	**0.0000**
Joint	**6**	**14.42**	**0.0000**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.7418696**	**.4164836**	**-1.78**	**0.085**	**-1.592443**	**.1087034**
(30 vs base) 1	**-.4758325**	**.446629**	**-1.07**	**0.295**	**-1.387971**	**.4363055**
(30 vs base) 2	**-.7845476**	**.4164836**	**-1.88**	**0.069**	**-1.635121**	**.0660254**
(74 vs base) 0	**-.438074**	**.4164836**	**-1.05**	**0.301**	**-1.288647**	**.4124991**
(74 vs base) 1	**2.219899**	**.4164836**	**5.33**	**0.000**	**1.369326**	**3.070472**
(74 vs base) 2	**1.954411**	**.446629**	**4.38**	**0.000**	**1.042273**	**2.866549**

28 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	**2**	**1.24**	**0.3027**
30	**2**	**1.62**	**0.2139**
74	**2**	**9.60**	**0.0006**
Joint	**6**	**6.50**	**0.0002**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	**1.327099**	**.9928863**	**1.34**	**0.191**	**-.7006451**	**3.354844**
(1 vs base) 30	**1.593136**	**1.048128**	**1.52**	**0.139**	**-.5474256**	**3.733698**
(1 vs base) 74	**3.985072**	**.9928863**	**4.01**	**0.000**	**1.957328**	**6.012816**
(2 vs base) 0	**.1213864**	**.682237**	**0.18**	**0.860**	**-1.271927**	**1.5147**
(2 vs base) 30	**.0787084**	**.682237**	**0.12**	**0.909**	**-1.314605**	**1.472022**
(2 vs base) 74	**2.513871**	**.6885639**	**3.65**	**0.001**	**1.107636**	**3.920106**

```
29 .
30 .
31 . anova ESP TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)
```

Number of obs = **52** R-squared = **0.9539**
Root MSE = **12.4476** Adj R-squared = **0.9216**

Source	Partial SS	df	MS	F	Prob>F
Model	**96171.157**	**21**	**4579.5789**	**29.56**	**0.0000**
TimeNumer~l	**59965.08**	**2**	**29982.54**	**193.51**	**0.0000**
GroupNume~l	**1074.0542**	**2**	**537.02709**	**3.47**	**0.0442**
TimeNumer~l#GroupNume~l	**20484.883**	**4**	**5121.2208**	**33.05**	**0.0000**
Sex	**159.81106**	**1**	**159.81106**	**1.03**	**0.3179**
weight	**1620.3084**	**12**	**135.0257**	**0.87**	**0.5827**
Residual	**4648.2841**	**30**	**154.9428**		
Total	**100819.44**	**51**	**1976.8518**		

```
32 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	**2**	**16.74**	**0.0000**
1	**2**	**171.45**	**0.0000**
2	**2**	**65.98**	**0.0000**
Joint	**6**	**84.72**	**0.0000**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-37.57618**	**7.186627**	**-5.23**	**0.000**	**-52.25323**	**-22.89914**
(30 vs base) 1	**-30.62126**	**7.706798**	**-3.97**	**0.000**	**-46.36064**	**-14.88188**
(30 vs base) 2	**-29.41388**	**7.186627**	**-4.09**	**0.000**	**-44.09093**	**-14.73683**
(74 vs base) 0	**-3.356995**	**7.186627**	**-0.47**	**0.644**	**-18.03404**	**11.32005**
(74 vs base) 1	**101.6718**	**7.186627**	**14.15**	**0.000**	**86.99471**	**116.3488**
(74 vs base) 2	**58.47318**	**7.706798**	**7.59**	**0.000**	**42.7338**	**74.21256**

33 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	2	0.15	0.8623
30	2	0.22	0.8067
74	2	22.17	0.0000
Joint	6	23.75	0.0000
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	4.665486	17.13273	0.27	0.787	-30.32422	39.65519
(1 vs base) 30	11.62041	18.08595	0.64	0.525	-25.31602	48.55684
(1 vs base) 74	109.6942	17.13273	6.40	0.000	74.70453	144.6839
(2 vs base) 0	-2.357238	11.77233	-0.20	0.843	-26.39954	21.68507
(2 vs base) 30	5.805063	11.77233	0.49	0.626	-18.23724	29.84737
(2 vs base) 74	59.47294	11.8815	5.01	0.000	35.20767	83.7382

34 .

35 . anova ESV TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)

Number of obs = **52**   R-squared = **0.8783**
Root MSE = **9.86986**   Adj R-squared = **0.7931**

Source	Partial SS	df	MS	F	Prob>F
Model	21090.122	21	1004.2915	10.31	0.0000
TimeNumer~l	12007.692	2	6003.8461	61.63	0.0000
GroupNume~l	103.64747	2	51.823737	0.53	0.5929
TimeNumer~l#GroupNume~l	642.28018	4	160.57005	1.65	0.1881
Sex	174.51669	1	174.51669	1.79	0.1908
weight	6582.4633	12	548.5386	5.63	0.0001
Residual	2922.4261	30	97.414202		
Total	24012.548	51	470.83427		

36 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	2	20.97	0.0000
1	2	23.23	0.0000
2	2	19.82	0.0000
Joint	6	21.34	0.0000
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	-36.68464	5.698368	-6.44	0.000	-48.32226	-25.04702
(30 vs base) 1	-36.59652	6.110819	-5.99	0.000	-49.07648	-24.11656
(30 vs base) 2	-30.4639	5.698368	-5.35	0.000	-42.10152	-18.82628
(74 vs base) 0	-14.86648	5.698368	-2.61	0.014	-26.5041	-3.228862
(74 vs base) 1	.4936301	5.698368	0.09	0.932	-11.14399	12.13125
(74 vs base) 2	2.74164	6.110819	0.45	0.657	-9.738317	15.2216

37 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	2	1.53	0.2319
30	2	0.54	0.5863
74	2	0.08	0.9270
Joint	6	1.33	0.2766
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	-10.08612	13.58476	-0.74	0.464	-37.82991	17.65767
(1 vs base) 30	-9.998005	14.34058	-0.70	0.491	-39.28537	19.28936
(1 vs base) 74	5.273994	13.58476	0.39	0.701	-22.46979	33.01778
(2 vs base) 0	-15.90417	9.33443	-1.70	0.099	-34.96762	3.159277
(2 vs base) 30	-9.683437	9.33443	-1.04	0.308	-28.74689	9.380013
(2 vs base) 74	1.70395	9.420996	0.18	0.858	-17.53629	20.94419

```
38 .
39 . anova EDP TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)

                         Number of obs =         52    R-squared     =  0.7335
                         Root MSE      =    2.16513    Adj R-squared =  0.5469

                  Source | Partial SS         df         MS        F    Prob>F
  -----------------------+----------------------------------------------------
                   Model |  387.06549         21    18.43169      3.93  0.0003
                         |
            TimeNumer~l  |  96.573307          2   48.286654     10.30  0.0004
            GroupNume~l  |  22.594004          2   11.297002      2.41  0.1070
TimeNumer~l#GroupNume~l  |  5.9805186          4   1.4951297      0.32  0.8630
                    Sex  |  1.5720893          1   1.5720893      0.34  0.5668
                 weight  |  267.50166         12   22.291805      4.76  0.0003
                         |
                Residual |  140.63394         30   4.6877979
  -----------------------+----------------------------------------------------
                   Total |  527.69943         51   10.347048

40 . contrast TimeNumerical@GroupNumerical , effect

  Contrasts of marginal linear predictions

  Margins: asbalanced

  ------------------------------------------------
                               |         df           F        P>F
  -----------------------------+------------------------------------
  TimeNumerical@GroupNumerical |
                            0  |          2        2.93     0.0686
                            1  |          2        5.31     0.0106
                            2  |          2        2.38     0.1095
                        Joint  |          6        3.54     0.0090
                               |
                  Denominator  |         30
  ------------------------------------------------

  ------------------------------------------------------------------------------------------------
                               |   Contrast   Std. err.       t    P>|t|     [95% conf. interval]
  -----------------------------+------------------------------------------------------------------
  TimeNumerical@GroupNumerical |
               (30 vs base) 0  |  -2.349526    1.25004    -1.88   0.070    -4.902448    .2033959
               (30 vs base) 1  |  -3.937982   1.340518    -2.94   0.006    -6.675686   -1.200279
               (30 vs base) 2  |  -1.974251    1.25004    -1.58   0.125    -4.527173    .5786704
               (74 vs base) 0  |   .4790139    1.25004     0.38   0.704    -2.073908    3.031936
               (74 vs base) 1  |  -.1510005    1.25004    -0.12   0.905    -2.703922    2.401921
               (74 vs base) 2  |   .8003607   1.340518     0.60   0.555    -1.937343    3.538064
  ------------------------------------------------------------------------------------------------

41 . contrast GroupNumerical@TimeNumerical , effect

  Contrasts of marginal linear predictions

  Margins: asbalanced
```

	df	F	P>F
GroupNumerical@TimeNumerical			
0	2	1.86	0.1725
30	2	2.26	0.1216
74	2	2.01	0.1512
Joint	6	0.97	0.4598
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	-4.905982	2.980063	-1.65	0.110	-10.99208	1.180118
(1 vs base) 30	-6.494438	3.145864	-2.06	0.048	-12.91915	-.0697264
(1 vs base) 74	-5.535997	2.980063	-1.86	0.073	-11.6221	.5501031
(2 vs base) 0	-3.717247	2.047675	-1.82	0.079	-7.899158	.4646645
(2 vs base) 30	-3.341972	2.047675	-1.63	0.113	-7.523883	.8399391
(2 vs base) 74	-3.3959	2.066665	-1.64	0.111	-7.616593	.8247934

42 .
43 .
44 . anova EDV TimeNumerical i.GroupNumerical TimeNumerical#GroupNumerical (Sex weight)

Number of obs = 52   R-squared = 0.9109
Root MSE = 11.6072   Adj R-squared = 0.8486

Source	Partial SS	df	MS	F	Prob>F
Model	41344.609	21	1968.7909	14.61	0.0000
TimeNumer~l	18489.415	2	9244.7076	68.62	0.0000
GroupNume~l	669.12253	2	334.56126	2.48	0.1005
TimeNumer~l#GroupNume~l	810.72778	4	202.68194	1.50	0.2259
Sex	619.55522	1	619.55522	4.60	0.0402
weight	16863.544	12	1405.2953	10.43	0.0000
Residual	4041.8133	30	134.72711		
Total	45386.422	51	889.92985		

45 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
TimeNumerical@GroupNumerical			
0	2	24.95	0.0000
1	2	30.93	0.0000
2	2	15.35	0.0000
Joint	6	23.74	0.0000
Denominator	30		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-42.67399**	**6.70142**	**-6.37**	**0.000**	**-56.36012**	**-28.98787**
(30 vs base) 1	**-44.3715**	**7.186473**	**-6.17**	**0.000**	**-59.04824**	**-29.69477**
(30 vs base) 2	**-34.17486**	**6.70142**	**-5.10**	**0.000**	**-47.86099**	**-20.48874**
(74 vs base) 0	**-3.591627**	**6.70142**	**-0.54**	**0.596**	**-17.27775**	**10.0945**
(74 vs base) 1	**9.592174**	**6.70142**	**1.43**	**0.163**	**-4.093952**	**23.2783**
(74 vs base) 2	**-3.314872**	**7.186473**	**-0.46**	**0.648**	**-17.99161**	**11.36186**

```
46 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	F	P>F
GroupNumerical@TimeNumerical			
0	**2**	**2.79**	**0.0774**
30	**2**	**2.87**	**0.0725**
74	**2**	**1.62**	**0.2140**
Joint	**6**	**1.83**	**0.1270**
Denominator	**30**		

	Contrast	Std. err.	t	P>|t|	[95% conf.	interval]
GroupNumerical@TimeNumerical						
(1 vs base) 0	**-37.48492**	**15.97601**	**-2.35**	**0.026**	**-70.1123**	**-4.857548**
(1 vs base) 30	**-39.18243**	**16.86487**	**-2.32**	**0.027**	**-73.6251**	**-4.73977**
(1 vs base) 74	**-24.30112**	**15.97601**	**-1.52**	**0.139**	**-56.92849**	**8.326252**
(2 vs base) 0	**-18.40145**	**10.97752**	**-1.68**	**0.104**	**-40.82053**	**4.017637**
(2 vs base) 30	**-9.902319**	**10.97752**	**-0.90**	**0.374**	**-32.3214**	**12.51676**
(2 vs base) 74	**-18.12469**	**11.07932**	**-1.64**	**0.112**	**-40.75169**	**4.502301**

```
47 .
   end of do-file

48 .
```